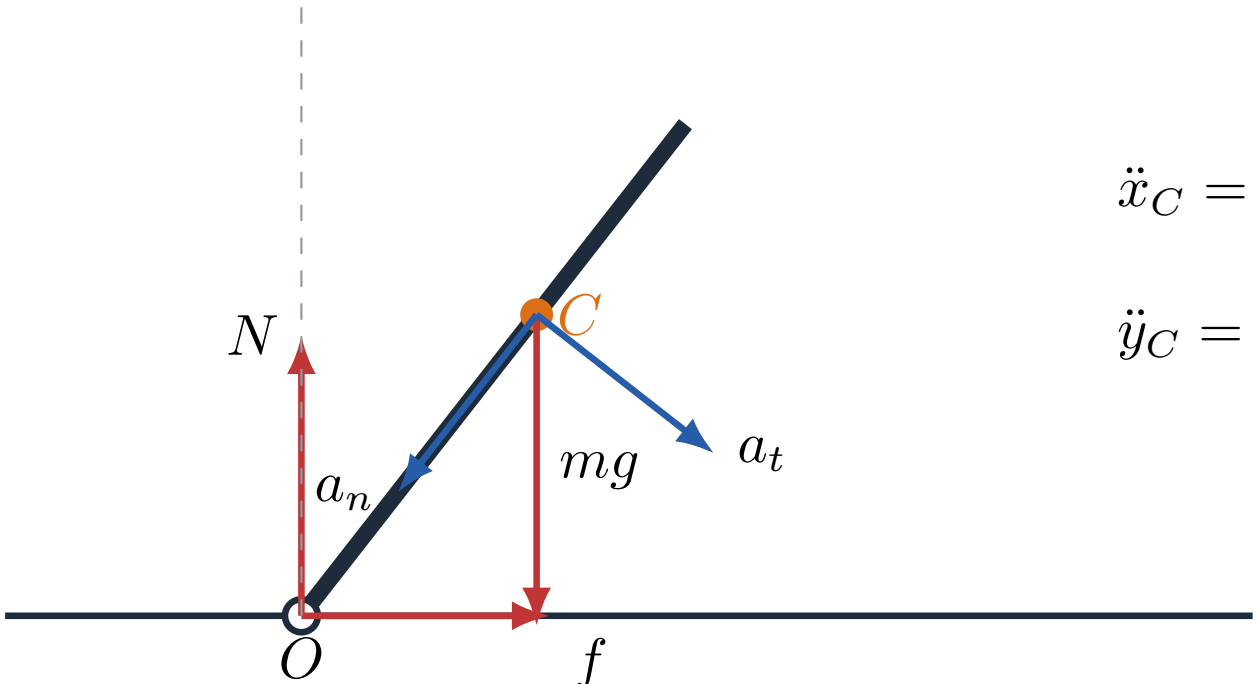

$$\ddot{x}_C = \frac{g}{\lambda}\sin\theta(3\cos\theta - 2)$$

$$\ddot{y}_C = -\frac{g}{\lambda}(1 + 2\cos\theta - 3\cos^2\theta)$$

$$a_t = r_c\ddot{\theta}, \quad a_n = r_c\dot{\theta}^2$$